Gap analysis — policy, strategy and standards

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Method. The jurist agent records a gap when it meets a situation that no binding text in the knowledge base settles, or that texts settle only in general terms. Gaps were found in two ways: **by trigger** (a rule or a question leads to a provision that is silent on the point) and **by absence** (no provision found at all). For each gap we list what UAE texts already say, what is missing, and global policy examples. In line with the hackathon brief, these are domain-specific findings with global examples, not recommendations addressed to one country.

ID	Domain	Gap	Detected in the scenarios by
L01	Liability and accountability	Who answers for an AI suggestion that a clinician validated?	Rule IA-SUGGESTION (doctor prescribes after an AI suggestion)
L02	Regulatory status of software	When is a decision-support or communication assistant a medical device?	Absence: no software classification criteria found
L03	Accessible health communication	No operational standard for meeting deaf patients' communication needs in care	Rule INFORMATION-ACCESSIBLE; advisor question in the crisis
L04	Informed consent	Is consent given through pictograms valid informed consent?	Rule CONSENTEMENT-PICTO; advisor question in the crisis
L05	Standards	No validation standard for medical pictograms	Rule ENVOI-DIAGNOSTIC (treatment explained in pictograms)
L06	Data governance and innovation	Strict health-data localisation versus cloud AI and international sandboxes	Rules HORS-EAU-SANS-EXCEPTION and ENTRAINEMENT-IA (foreign cloud AI)
L07	Human oversight and traceability	Oversight rules do not address multi-agent systems	Absence: no rule on agent-to-agent interaction
L08	Crisis coordination	No unified channel for AI incidents in health across emirates	Advisor question on an AI incident
L09	Accessible emergency communication	Emergency protocols do not guarantee equivalent access for deaf people	Rule URGENCE-ACCESSIBLE (emergency of a deaf patient)

L01 · Who answers for an AI suggestion that a clinician validated?

Situation. The assistant agent proposes an alternative treatment; the doctor accepts it; it later proves wrong.

What UAE texts already say.

- Defines the practitioner's medical error; silent on AI-generated suggestions. — Federal Decree-Law No. (4) of 2016 Concerning Medical Liability, Art. 6 (official text) https://uaelegislation.gov.ae/en/legislations/1192
- A human makes the final decision and is responsible for actions taken by the technology. — Department of Health Abu Dhabi - Policy on Use of Artificial Intelligence in the Healthcare Sector (Policy/AI/0.9), Policy statement (AI as supportive technology) (official text) https://www.doh.gov.ae/-/media/E9C1470A575146B18015DEBE57E47F8D.ashx
- Accountability for outcomes must be agreed; developers accountable to end users; appeals procedure. — Dubai Health Authority - Policy: Artificial Intelligence in the Healthcare (DHA/HISHD/PP-09), Sec. 4.2.1-4.2.3 (official text) https://www.dha.gov.ae/uploads/082022/Artificial%20Intelligence%20Policy20228457.pdf

What is missing. No binding rule allocating liability between clinician, software developer and facility when a validated AI suggestion causes harm, and no rule on the evidential value of AI logs in a dispute.

Global examples.

- Software is a product; disclosure of evidence and presumption of defect ease the claimant's burden. — Directive (EU) 2024/2853 on liability for defective products, Arts. 4, 9, 10 (official text) https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX%3A32024L2853
- Oversight duties include awareness of automation bias and the ability to override. — Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act), Art. 14 (official text) https://ai-act-service-desk.ec.europa.eu/en/ai-act/article-12

How MEDIMCOM IA handles it today. Every AI suggestion carries its author agent, sources and status; overriding an A_REVOIR verdict requires a named doctor; the hash-chained journal can be produced as evidence.

L02 · When is a decision-support or communication assistant a medical device?

Situation. MEDIMCOM translates, suggests and checks; none of its agents prescribes, but its outputs influence decisions.

What UAE texts already say.

- Medical equipment includes operating software and AI-based products. — Federal Decree-Law No. (38) of 2024 Governing Medical Products, Pharmacists and Pharmaceutical Establishments, Art. 1 (definition of Medical Equipment) (official text) https://uaelegislation.gov.ae/en/legislations/2751
- Risk classes; high-risk systems need conformity assessment (Abu Dhabi). — Department of Health Abu Dhabi - Responsible Artificial Intelligence (AI) Standard (DoH/ST/DDGO/RAI/V1/2025), Sec. 3.3.1.2.2 (official text) https://www.doh.gov.ae/-/media/B84FF5DD027846699C51F36C439A580F.ashx

What is missing. The decree-law text read does not set criteria separating regulated decision-support software from non-device tools (communication aids, compliance checkers); executive regulations or EDE guidance on software to be checked.

Global examples.

- Four criteria; the clinician must be able to review the basis of each recommendation. — FDA guidance: Clinical Decision Support Software (revised final guidance), Four non-device criteria (official text) https://www.fda.gov/regulatory-information/search-fda-guidance-documents/clinical-decision-support-software
- Decision-support software is at least class IIa. — Regulation (EU) 2017/745 on medical devices - Annex VIII, Rule 11, Annex VIII, Rule 11 (secondary source) https://www.bristows.com/news/the-regulation-of-software-medical-devices
- Regulatory sandbox to test AI medical devices before rules are final. — MHRA AI Airlock - regulatory sandbox for AI as a Medical Device, Programme description (official text) https://www.gov.uk/government/news/mhra-launches-ai-airlock-to-address-challenges-for-regulating-medical-devices-that-use-artificial-intelligence

How MEDIMCOM IA handles it today. Designed to meet the 'reviewable basis' test: every suggestion and verdict shows its source text; the doctor decides.

L03 · No operational standard for meeting deaf patients' communication needs in care

Situation. A deaf patient with limited literacy consults without a sign-language interpreter.

What UAE texts already say.

- Right to information and expression through sign language and other means. — Federal Law No. (29) of 2006 Concerning the Rights of People with Special Needs (People of Determination), as amended by Federal Law No. (14) of 2009, Art. 7 (official text) https://elaws.moj.gov.ae/UAE-MOJ_LC-En/00_SOCIAL%20WELFARE/UAE-LC-En_2006-08-13_00029_Kait.html
- Telehealth must make provision for People of Determination; language easily understood (Dubai). — Dubai Health Authority - Standards for Telehealth Services (DHA/HRS/HPSD/ST-14, version 4), Sec. 14.1.5(a), 8.3.5, 14.1.6, 9.6.3 (official text) https://dha.gov.ae/uploads/012023/Standards%20for%20Telehealth%20Services2023158613.pdf
- Accept and facilitate augmentative and alternative communication in official interactions. — Convention on the Rights of Persons with Disabilities, Art. 21(a)-(b),(e) (official text) https://social.desa.un.org/issues/disability/crpd/convention-on-the-rights-of-persons-with-disabilities-articles

What is missing. No federal operational standard requiring providers to identify, record and flag a patient's communication need, nor rules on which alternative means (pictograms, easy-read, sign avatars) are acceptable in a clinical encounter.

Global examples.

- Identify, record, flag, share and meet communication needs - a legal requirement in England since 2016. — NHS England - Accessible Information Standard (DCB1605), Standard DCB1605 (official text) https://www.england.nhs.uk/accessibleinfo

How MEDIMCOM IA handles it today. The communication need is stored in the patient state and travels with the record; the translator agent adapts every message; the doctor validates each pictogram sequence.

L04 · Is consent given through pictograms valid informed consent?

Situation. The patient agrees to a treatment by selecting pictograms.

What UAE texts already say.

- No treatment without consent, except emergencies. — Federal Decree-Law No. (4) of 2016 Concerning Medical Liability, Art. 5(1) (official text) https://uaelegislation.gov.ae/en/legislations/1192
- Written consent for surgery. — Federal Decree-Law No. (4) of 2016 Concerning Medical Liability, Art. 8(1)(c) (official text) https://uaelegislation.gov.ae/en/legislations/1192
- Consent documented at each encounter; electronic consent accepted (Dubai telehealth). — Dubai Health Authority - Standards for Telehealth Services (DHA/HRS/HPSD/ST-14, version 4), Sec. 10.1, 10.1.1, 10.4.2 (official text) https://dha.gov.ae/uploads/012023/Standards%20for%20Telehealth%20Services2023158613.pdf

What is missing. No provision recognises consent expressed through augmentative and alternative communication, defines how comprehension is checked, or how such consent is documented.

Global examples.

- Support in exercising legal capacity. — Convention on the Rights of Persons with Disabilities, Art. 12 (official text) https://social.desa.un.org/issues/disability/crpd/convention-on-the-rights-of-persons-with-disabilities-articles
- Care of the same quality on the basis of free and informed consent. — Convention on the Rights of Persons with Disabilities, Art. 25 (official text) https://social.desa.un.org/issues/disability/crpd/convention-on-the-rights-of-persons-with-disabilities-articles
- Alternative communication accepted in official interactions. — Convention on the Rights of Persons with Disabilities, Art. 21(a)-(b),(e) (official text) https://social.desa.un.org/issues/disability/crpd/convention-on-the-rights-of-persons-with-disabilities-articles

How MEDIMCOM IA handles it today. Consent is recorded with its means of expression, a comprehension check (teach-back with pictograms) and the doctor's attestation, all in the journal.

L05 · No validation standard for medical pictograms

Situation. Pictograms are generated or chosen by a model and shown to patients.

What UAE texts already say.

- No provision found in the knowledge base.

What is missing. No UAE standard found on testing the comprehensibility of health pictograms or of AI-generated images used to inform patients.

Global examples.

- Standard method for testing what proportion of people understand a symbol. — ISO 9186-1:2014 Graphical symbols - Test methods - Part 1: Method for testing comprehensibility, Scope (Introduction) (official text) https://webstore.ansi.org/preview-pages/ISO/preview_ISO+9186-1-2014.pdf

How MEDIMCOM IA handles it today. Pictogram library tested with deaf users following the ISO 9186-1 method before use; every sequence validated by the doctor.

L06 · Strict health-data localisation versus cloud AI and international sandboxes

Situation. A model or vector store hosted outside the UAE (e.g. an international sandbox) would process consultation content.

What UAE texts already say.

- No storage or processing of health data outside the State, except by resolution. — Federal Law No. (2) of 2019 Concerning the Use of Information and Communication Technology in Health Fields, Art. 13 (official text) https://uaelegislation.gov.ae/en/legislations/1209
- Listed exceptions (research, foreign labs, insurers, etc.) - none explicitly for AI inference or model training. — Ministry of Health and Prevention Resolution No. (51) of 2021 on cases where health data may be stored or transferred outside the State, list of exceptions (secondary source) https://mohap.gov.ae/en/w/ministerial-decree-no.-51-of-the-year-2021-regarding-cases-in-which-health-data-and-information-may-be-stored-or-transferred-outside-the-country
- Compute and storage in the UAE (Abu Dhabi). — Department of Health Abu Dhabi - Responsible Artificial Intelligence (AI) Standard (DoH/ST/DDGO/RAI/V1/2025), Sec. 3.1.4.5.1 (official text) https://www.doh.gov.ae/-/media/B84FF5DD027846699C51F36C439A580F.ashx

What is missing. No explicit pathway for AI testing in cross-border or international sandboxes, and no statement on whether synthetic or de-identified data falls outside the localisation rule.

Global examples.

- Sandboxes with strict conditions; anonymised or synthetic data preferred. — Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act), Arts. 57-59 (official text) https://ai-act-service-desk.ec.europa.eu/en/ai-act/article-12
- National sandbox for AI medical devices. — MHRA AI Airlock - regulatory sandbox for AI as a Medical Device, Programme description (official text) https://www.gov.uk/government/news/mhra-launches-ai-airlock-to-address-challenges-for-regulating-medical-devices-that-use-artificial-intelligence

How MEDIMCOM IA handles it today. Local models only; synthetic data in the prototype; the jurist blocks any outbound transfer without an exception and its conditions.

L07 · Oversight rules do not address multi-agent systems

Situation. Several AI agents exchange messages and act on a shared record before a human sees the result.

What UAE texts already say.

- All decisions logged and traceable to a responsible party. — Department of Health Abu Dhabi - Responsible Artificial Intelligence (AI) Standard (DoH/ST/DDGO/RAI/V1/2025), Sec. 3.1.7.1.3 (official text) https://www.doh.gov.ae/-/media/B84FF5DD027846699C51F36C439A580F.ashx
- AI supervised by end users and can be overridden. — Dubai Health Authority - Policy: Artificial Intelligence in the Healthcare (DHA/HISHD/PP-09), Sec. 4.4.1-4.4.2 (official text) https://www.dha.gov.ae/uploads/082022/Artificial%20Intelligence%20Policy20228457.pdf
- Failures traced, diagnosed and controlled. — Dubai Health Authority - Policy: Artificial Intelligence in the Healthcare (DHA/HISHD/PP-09), Sec. 4.3.1 (official text) https://www.dha.gov.ae/uploads/082022/Artificial%20Intelligence%20Policy20228457.pdf

What is missing. No requirement on agent-to-agent communication, limits of agent autonomy, tool permissions, or tamper-evident logs.

Global examples.

- Automatic logging over the lifetime of the system. — Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act), Art. 12 (official text) https://ai-act-service-desk.ec.europa.eu/en/ai-act/article-12
- Effective human oversight including automation bias. — Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act), Art. 14 (official text) https://ai-act-service-desk.ec.europa.eu/en/ai-act/article-12

How MEDIMCOM IA handles it today. Typed messages between agents, per-agent tool permissions, deterministic verdicts, SHA-256 chained journal.

L08 · No unified channel for AI incidents in health across emirates

Situation. During a crisis an AI component fails or gives harmful output in several facilities.

What UAE texts already say.

- Report AI incidents to DHA (Dubai). — Dubai Health Authority - Policy: Artificial Intelligence in the Healthcare (DHA/HISHD/PP-09), Sec. 4.6.17, 4.8.1-4.8.2, 4.8.5 (official text) https://www.dha.gov.ae/uploads/082022/Artificial%20Intelligence%20Policy20228457.pdf
- Report adverse outcomes to governance bodies (Abu Dhabi). — Department of Health Abu Dhabi - Responsible Artificial Intelligence (AI) Standard (DoH/ST/DDGO/RAI/V1/2025), Sec. 3.1.7.1.2.3 (official text) https://www.doh.gov.ae/-/media/B84FF5DD027846699C51F36C439A580F.ashx
- 24-hour notification model exists for communicable diseases. — Federal Law No. (14) of 2014 On Control of Communicable Diseases, Art. 4 (official text) https://uaelegislation.gov.ae/en/legislations/1198

What is missing. Incident reporting duties are emirate-specific; no federal time limit or shared format for AI incidents found in the knowledge base.

Global examples.

- Serious incidents reported within 15 days, shorter for widespread or fatal ones. — Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act), Art. 73 (official text) https://ai-act-service-desk.ec.europa.eu/en/ai-act/article-12
- Time-bound notification through a single national focal point. — International Health Regulations (2005), Art. 6(1) (official text) https://apps.who.int/gb/bd/pdf_files/IHR_2022-en.pdf

How MEDIMCOM IA handles it today. The journal exports an incident record (what, when, which agent, which sources); the jurist reminds the time limits that do exist.

L09 · Emergency protocols do not guarantee equivalent access for deaf people

Situation. In an emergency the usual voice channels and interpreters are unavailable.

What UAE texts already say.

- Tested emergency protocols in telehealth, no accessibility requirement. — Dubai Health Authority - Standards for Telehealth Services (DHA/HRS/HPSD/ST-14, version 4), Sec. 12.4-12.4.3, 14.1.4(c) (official text) https://dha.gov.ae/uploads/012023/Standards%20for%20Telehealth%20Services2023158613.pdf
- General right to communicate through other means. — Federal Law No. (29) of 2006 Concerning the Rights of People with Special Needs (People of Determination), as amended by Federal Law No. (14) of 2009, Art. 7 (official text) https://elaws.moj.gov.ae/UAE-MOJ_LC-En/00_SOCIAL%20WELFARE/UAE-LC-En_2006-08-13_00029_Kait.html
- Protection in situations of risk. — Convention on the Rights of Persons with Disabilities, Art. 11 (official text) https://social.desa.un.org/issues/disability/crpd/convention-on-the-rights-of-persons-with-disabilities-articles

What is missing. No requirement found that health emergency protocols provide equivalent non-voice access for deaf patients.

Global examples.

- Equivalent access to emergency services for end-users with disabilities. — Directive (EU) 2018/1972 establishing the European Electronic Communications Code - Article 109, Art. 109(5) (secondary source) https://www.legislation.gov.uk/eudr/2018/1972/article/109

How MEDIMCOM IA handles it today. Emergency pictogram set always available offline; urgent alerts are prioritised by the assistant agent.